THAMARIKANNAN K R

Electronics and Communication Engineering Student

Embedded Systems | Digital Electronics | Python | Java

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ABOUT ME

Motivated Electronics and Communication Engineering student with strong fundamentals in embedded systems, digital electronics, and IoT-based communication. Skilled in programming, circuit analysis, and mini-project development, with hands-on experience building sensor-driven assistive devices. Eager to apply technical knowledge and contribute effectively in academic and industry environments.

EDUCATION

B.E. Electronics and Communication Engineering
Velalar College of Engineering and Technology, Erode
CGPA: 7.3 | 2023 - 2027

Class XII — Computer Science
Government Model Higher Secondary School, Kavindapadi
66% | Jun 2022 - May 2023

Class X
Government Higher Secondary School, Polavakalipalayam
Pass | Jun 2020 - May 2021

TECHNICAL SKILLS

Programming:

- Python
- Java (Beginner)
- C (Embedded)

Embedded Platforms:

- Arduino
- ESP32

Technologies:

- IoT Systems
- Sensor Interfacing
- Circuit Analysis
- GSM/Wi-Fi Communication

AREAS OF INTEREST

- Embedded Systems
- IoT Applications
- Digital Electronics
- Assistive Technology
- Software Development

INTERNSHIP & TECHNICAL TRAINING

IMPLANT TRAINING
Ashwin Builders

- Completed hands-on implant training covering practical construction and industrial site exposure.

TECHNICAL WORKSHOP — EMBEDDED SYSTEMS AND IOT
PSG Tech

- Attended a technical workshop on embedded systems and IoT applications, strengthening core electronics and communication engineering concepts.

CERTIFIED IN EMBEDDED SYSTEMS
Santech Technology

- Completed hands-on certification covering microcontroller programming and embedded system design.

WORKSHOP — NEURAL NETWORKS
National Institute of Technology, Tiruchirappalli

- Attended a hands-on workshop covering neural network fundamentals, architecture design, training techniques, and real-world AI applications.

SOLAR PANEL INSTALLATION — HANDS-ON TRAINING

- Completed hands-on training in solar panel installation, covering panel mounting, wiring, and basic system setup for renewable energy applications.

ACADEMIC PROJECTS

SMART CRUTCH FOR FALL DETECTION AND EMERGENCY COMMUNICATION USING IOT

- Designed a smart crutch fitted with accelerometer and gyroscope sensors to detect sudden falls in real time.
- Used an IoT module to automatically send the user's live location and an emergency alert to registered contacts and caregivers upon fall detection.
- Integrated a microcontroller (Arduino/ESP32) to process sensor data and trigger GSM/Wi-Fi based communication without manual intervention.
- Aimed at improving safety and independence for elderly and mobility-impaired individuals through low-cost, wearable assistive technology.

CERTIFICATIONS

- Introduction to Python — Infosys (Jun 2025)
- Certified in Embedded Systems — Santech Technology (Oct 2024)
- Project Exhibition Certificate — Smart Crutch for Fall Detection and Emergency Communication using IoT

LEADERSHIP EXPERIENCE

PAPER PRESENTATION — KPR INSTITUTE OF ENGINEERING AND TECHNOLOGY

- Presented a technical paper on the Smart Crutch fall detection and emergency communication system, showcasing IoT-based assistive technology research to a multi-college audience.